

Nama: Adi Darmiana

NIM: 310125023966

4 digit NIM: 0,3966 (multiplier)

Learning rate: 1.0

Konstanta e : 2.71828183

Data training sebelum skalasi

Data 1: $x_1 = 0,9; x_2 = 0,4; t = 1$

Data 2: $x_1 = 0,73; x_2 = 0,85; t = 0$

Data training setelah skalasi ($\times 0,3966$)

Data 1: $x_1 = 0,356940; x_2 = 0,158640; t = 1$

Data 2: $x_1 = 0,289518; x_2 = 0,337110; t = 0$

Bobot awal (acak) sebelum skalasi

$v_{11} = 0,9562 \quad v_{12} = 0,7762 \quad v_{13} = 0,1623 \quad v_{14} = 0,2886$

$v_{21} = 0,1962 \quad v_{22} = 0,6133 \quad v_{23} = 0,0311 \quad v_{24} = 0,9711$

$v_{01} = 0,7496 \quad v_{02} = 0,3796 \quad v_{03} = 0,7256 \quad v_{04} = 0,1628$

$w_1 = 0,7280 \quad w_2 = 0,9585 \quad w_3 = 0,6999 \quad w_4 = 0,0550$

$w_0 = 0,9505$

Bobot awal (acak) setelah skalasi

$v_{11} = 0,379229 \quad v_{12} = 0,307841 \quad v_{13} = 0,064368 \quad v_{14} = 0,114459$

$v_{21} = 0,077813 \quad v_{22} = 0,243235 \quad v_{23} = 0,012334 \quad v_{24} = 0,385138$

$$v_{01} = 0,297291 \quad v_{02} = 0,150549 \quad v_{03} = 0,287773 \quad v_{04} = 0,064566$$

$$w_1 = 0,090425 \quad w_2 = 0,380141 \quad w_3 = 0,269648 \quad w_4 = 0,021813$$

$$w_0 = 0,376968$$

EPOCH 1 DATA 1

Operasi pada Hidden Layer

Rumus Penjumlahan terbobot ($z_{in i}$)

$$z_{in i} = v_{0i} + (v_{1i} * x_{1i}) + (v_{2i} * x_{2i})$$

$$z_{in 1} = 0,297291 + (0,376968 * 0,356940) + (0,07783 * 0,158640)$$

$$= 0,297291 + 0,135362 + 0,012344 = 0,444997$$

$$z_{in 2} = 0,150549 + (0,380141 * 0,356940) + (0,243235 * 0,158640)$$

$$= 0,150549 + 0,135362 + 0,038587 = 0,299017$$

$$z_{in 3} = 0,287773 + (0,064368 * 0,356940) + (0,012334 * 0,158640)$$

$$= 0,287773 + 0,022976 + 0,001957 = 0,312705$$

$$z_{in 4} = 0,064566 + (0,114459 * 0,356940) + (0,385138 * 0,158640)$$

$$= 0,064566 + 0,040855 + 0,061098 = 0,166520$$

Rumus Pengaktifan (z)

$$z_i = \frac{1}{1 + e^{-z_{in i}}}$$

$$z_1 = \frac{1}{1 + 2,71828183^{-0,444998}}$$

$$= \frac{1}{0,609450} = 0,609449$$

$$z_2 = \frac{1}{1 + 2,71828183^{-0,299017}}$$

$$= \frac{1}{1,741547} = 0,574202$$

$$z_3 = \frac{1}{1 + 2,71828183^{-0,312106}}$$

$$= \frac{1}{1,731466} = 0,577545$$

$$z_4 = \frac{1}{1 + 2,71828183^{-0,166520}}$$

$$= \frac{1}{1,846605} = 0,541534$$

⇒ Operasi pada output layer

Rumus perkalian linier combiner

$$y_{in} = w_0 + (w_1 * z_1) + (w_2 * z_2) + (w_3 * z_3) + (w_4 * z_4)$$

$$y_{in} = 0,376968 + (0,090425 * 0,609449) + (0,380141 * 0,574202)$$

$$+ (0,269648 * 0,577545) + (0,621813 * 0,541534)$$

$$= 0,817902$$

Rumus pengaktifan (y)

$$y = \frac{1}{1 + e^{-y - \ln}}$$

$$y = \frac{1}{1 + 2,71828183^{-0,817902}} = \frac{1}{1 + 0,441355}$$
$$= \frac{1}{1,441355} = 0,693791$$

Rumus error

$$\text{error} = t - y$$

$$\text{error} = 1 - 0,693791 = 0,306209$$

$$\text{kuadrat error} = (0,306209)^2 = 0,093764$$

Rumus menghitung nilai δ

$$\delta = (t_1 - y) * \left(\frac{1}{1 + e^{-y - \ln}} \right) * \left[1 - \left(\frac{1}{1 + e^{-y - \ln}} \right) \right]$$

$$\delta = (1 - 0,693791) * \left(\frac{1}{1,441355} \right) * \left[1 - \left(\frac{1}{1,441355} \right) \right]$$

$$= 0,306209 * 0,693791 * [1 - 0,693791]$$

$$= 0,306209 * 0,693791 * 0,306209$$

$$= 0,306209 * 0,212445$$

$$= 0,065053$$

Menghitung Δw rumus

$$\Delta w_i = a * f * z_i \quad | \quad \Delta w_0 = a * f$$

$$\Delta w_1 = 1.0 * 0,065053 * 0,609449 = 0,039646$$

$$\Delta w_2 = 1.0 * 0,065053 * 0,54122 = 0,037353$$

$$\Delta w_3 = 1.0 * 0,065053 * 0,577545 = 0,037571$$

$$\Delta w_4 = 1.0 * 0,065053 * 0,541534 = 0,035228$$

$$\Delta w_0 = 1.0 * 0,065053 = 0,065053$$

Rumus w_i

$$w_i' = w_i + \Delta w_i$$

$$w_1' = 0,090425 + 0,039646 = 0,130071$$

$$w_2' = 0,380141 + 0,037353 = 0,417494$$

$$w_3' = 0,269648 + 0,037571 = 0,307219$$

$$w_4' = 0,04813 + 0,035228 = 0,083358$$

$$w_0' = 0,376968 + 0,065053 = 0,442021$$

Rumus menghitung (f_i)

$$f_i = s_{ini} * \left(\frac{1}{1 + e^{-2f_i}} \right) * \left[1 - \left(\frac{1}{1 + e^{2f_i}} \right) \right]$$

$$f_1 = 0,005202 * \left(\frac{1}{1 + 2,71828183^{0,130071}} \right)$$

$$* \left[1 - \left(\frac{1}{1 + 2,71828183^{0,130071}} \right) \right]$$

→ rumus (w)

$$s_i \cdot \bar{w}_i = f * w_i$$

$$s_i \cdot \bar{w}_1 = 0,065053 * 0,90428 = 0,005882$$

$$s_i \cdot \bar{w}_2 = 0,065053 * 0,380141 = 0,024729$$

$$s_i \cdot \bar{w}_3 = 0,065053 * 0,269648 = 0,017541$$

$$s_i \cdot \bar{w}_4 = 0,065053 * 0,021813 = 0,001419$$

$$\begin{aligned} \rightarrow s_1 &= 0,005882 * 0,609449 * 0,390550 \\ &= 0,005882 * 0,238020 \\ &= 0,001400 \end{aligned}$$

$$\begin{aligned} s_2 &= 0,024729 * 0,574202 * 0,425798 \\ &= 0,024729 * 0,2444946 \\ &= 0,006046 \end{aligned}$$

$$\begin{aligned} s_3 &= 0,017541 * 0,577545 * 0,422455 \\ &= 0,017541 * 0,244036 \\ &= 0,004280 \end{aligned}$$

$$\begin{aligned} s_4 &= 0,001419 * 0,541534 * 0,458466 \\ &= 0,001419 * 0,248275 \\ &= 0,000352 \end{aligned}$$

Menghitung ΔV_1 , ΔV_2 , dan ΔV_0

$$\Delta V_{11} = a * f_1 * X_1$$

$$= 1 * 0,001400 * 0,356940$$

$$= 0,000500$$

$$\Delta V_{12} = a * f_2 * X_1$$

$$= 1 * 0,006046 * 0,356940$$

$$= 0,002158$$

$$\Delta V_{13} = a * f_3 * X_1$$

$$= 1 * 0,004280 * 0,356940$$

$$= 0,001528$$

$$\Delta V_{14} = a * f_4 * X_1$$

$$= 1 * 0,000352 * 0,356940$$

$$= 0,000126$$

$$\Delta V_{21} = a * f_1 * X_2$$

$$= 1 * 0,001400 * 0,158640$$

$$= 0,000222$$

$$\Delta V_{22} = a * f_2 * X_2$$

$$= 1 * 0,006046 * 0,158640$$

$$= 0,000959$$

$$\Delta V_{23} = a * f_3 * X_2$$

$$= 1 * 0,004280 * 0,158640$$

$$= 0,000679$$

$$\Delta V_{24} = a * f_4 * X_2$$

$$= 1 * 0,000352 * 0,158640$$

$$= 0,000056$$

$$\Delta v_1 = a * f_1$$

$$= 1 * 0,001400 = 0,001400$$

$$\Delta v_2 = a * f_2$$

$$= 1 * 0,006046 = 0,006046$$

$$\Delta v_3 = a * f_3$$

$$= 1 * 0,004280 = 0,004280$$

$$\Delta v_4 = a * f_4$$

$$= 1 * 0,000352 = 0,000352$$

$\Rightarrow$ Perhitungan Bobot Boru (V_1 , V_2 , dan V_0)

$$V_1 = V_{11} + \Delta V_{11}$$

$$= 0,379229 + 0,000500$$

$$= 0,379729$$

$$V_2 = V_{12} + \Delta V_{12}$$

$$= 0,307841 + 0,002158$$

$$= 0,309999$$

$$V_3 = V_{13} + \Delta V_{13}$$

$$= 0,064368 + 0,001528$$

$$= 0,065896$$

$$V_4 = V_{14} + \Delta V_{14}$$

$$= 0,114459 + 0,000126$$

$$= 0,114585$$

$$V_{21} = V_{21} + \Delta V_{21}$$

$$= 0,077813 + 0,000222$$

$$= 0,078035$$

$$V_{22} = V_{22} + \Delta V_{22}$$

$$= 0,243235 + 0,000959$$

$$= 0,244194$$

$$V_{23} = 0,012334 + 0,000679$$

$$= 0,013013$$

$$V_{24} = V_{24} + \Delta V_{24}$$

$$= 0,385138 + 0,000056$$

$$= 0,385194$$

$$V_{01} = V_{01} + \Delta V_{01}$$

$$= 0,297291 + 0,001400$$

$$= 0,298691$$

$$V_{02} = V_{02} + \Delta V_{02}$$

$$= 0,150549 + 0,006046$$

$$= 0,156595$$

$$V_{03} = V_{03} + \Delta V_{03}$$

$$= 0,287773 + 0,004280$$

$$= 0,292053$$

$$V_{04} = V_{04} + \Delta V_{04}$$

$$= 0,064566 + 0,000352$$

$$= 0,064919$$

Epoch 1 Part 2.

⇒ Operasi pada Hidden Layer

$$\begin{aligned} z_{in1} &= V_{01} + (V_{11} * X_{12}) + (V_{21} * X_{22}) \\ &= 0,298691 + (0,379729 * 0,289518) + \\ &\quad (0,078035 * 0,337110) \\ &= 0,298691 + 0,109938 + 0,026306 = 0,434936 \end{aligned}$$

$$\begin{aligned} z_{in2} &= V_{02} + (V_{12} * X_{12}) + (V_{22} * X_{22}) \\ &= 0,156595 + (0,309999 * 0,289518) + \\ &\quad (0,244194 * 0,3377110) \\ &= 0,156595 + 0,089750 + 0,082320 = 0,328666 \end{aligned}$$

$$\begin{aligned} z_{in3} &= V_{03} + (V_{13} * X_{12}) + (V_{23} * X_{22}) \\ &= 0,292053 + (0,065896 * 0,289518) + \\ &\quad (0,013013 * 0,3377110) \\ &= 0,292053 + 0,019078 + 0,004387 = 0,315518 \end{aligned}$$

$$\begin{aligned} z_{in4} &= V_{04} + (V_{14} * X_{12}) + (V_{24} * X_{22}) \\ &= 0,064919 + (0,114585 * 0,289518) + \\ &\quad (0,385194 * 0,3377110) \\ &= 0,064919 + 0,033174 + 0,129853 = 0,227946 \end{aligned}$$

$$\begin{aligned} z_1 &= \frac{1}{1 + e^{-2z_{in1}}} = \frac{1}{1 + 2,71828183^{-0,434936}} \\ &= \frac{1}{1 + 0,647306} = 0,607052 \end{aligned}$$

$$z_2 = \frac{1}{1 + e^{-2z_{in2}}} = \frac{1}{1 + 2,71828183^{-0,328666}}$$

$$= \frac{1}{1 + 0,719885} = 0,581435$$

$$z_3 = \frac{1}{1 + e^{-2u_3}} = \frac{1}{1 + 2,71828183^{-0,315518}}$$

$$= \frac{1}{1 + 0,729411} = 0,578283$$

$$z_4 = \frac{1}{1 + e^{2u_4}} = \frac{1}{1 + 2,71828183^{-0,227946}}$$

$$= \frac{1}{1 + 0,786166} = 0,556741$$

($\Rightarrow$) Operasi pada output layer

$$y_{in} = w_0 + (w_1 * z_1) + (w_2 * z_2) + (w_3 * z_3) + (w_4 * z_4)$$

$$= 0,442021 + (0,130071 * 0,607052) + (0,417494 * 0,581435) + (0,307219 * 0,578283) + (0,057041 * 0,556741)$$

$$= 0,442021 + 0,078960 + 0,242746 + 0,177644 + 0,031757 = 0,973128$$

Pengaktifan y

$$y = \frac{1}{1 + e^{-y_{in}}} = \frac{1}{1 + 2,71828183^{-0,973128}}$$

$$= \frac{1}{1 + 0,377897} = 0,725744$$

$$\begin{aligned} \text{error} &= 0 - y \rightarrow \text{error} = t_2 - y \\ &= 0 - 0.725744 \\ &= -0.725744 \end{aligned}$$

$$\begin{aligned} (\text{error})^2 / \text{MSE} &= (-0.725744)^2 \\ &= 0.526762 \end{aligned}$$

$$f = (t_2 - y) * \left(\frac{1}{1 + e^{-y - t_2}} \right) * \left[1 - \left(\frac{1}{1 + e^{y + t_2}} \right) \right]$$

atau

$$\begin{aligned} f &= \text{error} * y * (1 - y) \\ &= -0.725744 * 0.725744 * (1 - 0.725744) \\ &= -0.725744 * 0.725744 * 0.274256 \\ &= -0.725744 * 0.199040 = -0.144452 \end{aligned}$$

$$\begin{aligned} \Delta w_1 &= 1 * f * z_1 \\ &= 1 * -0.144452 * 0.607052 = -0.087690 \end{aligned}$$

$$\begin{aligned} \Delta w_2 &= 1 * f * z_2 \\ &= 1 * -0.144452 * 0.581435 = -0.083989 \end{aligned}$$

$$\begin{aligned} \Delta w_3 &= 1 * f * z_3 \\ &= 1 * -0.144452 * 0.578232 = -0.083527 \end{aligned}$$

$$\begin{aligned} \Delta w_4 &= 1 * f * z_4 \\ &= 1 * -0.144452 * 0.556741 = -0.080422 \end{aligned}$$

$$\begin{aligned} \Delta w_0 &= 1 * f \\ &= 1 * -0.144452 = -0.144452 \end{aligned}$$

$$w_1 = w_1 + \Delta w_1$$

$$= 0,130071 + (-0,087690) = 0,042381$$

$$w_2 = w_2 + \Delta w_2$$

$$= 0,417494 + (-0,083989) = 0,333505$$

$$w_3 = w_3 + \Delta w_3$$

$$= 0,307219 + (-0,083527) = 0,223692$$

$$w_4 = w_4 + \Delta w_4$$

$$= 0,057041 + (-0,080422) = 0,023381$$

$$w_0 = w_0 + \Delta w_0$$

$$= 0,442021 + (-0,144452) = 0,297569$$

$$f_{in1} = f * w_1$$

$$= -0,144452 * 0,130071 = -0,018789$$

$$f_{in2} = f * w_2$$

$$= -0,144452 * 0,417494 = -0,060300$$

$$f_{in3} = f * w_3$$

$$= -0,144452 * 0,307219 = -0,044378$$

$$f_{in4} = f * w_4$$

$$= -0,144452 * 0,057041 = -0,008240$$

$$S_i = S_{in i} * \left(\frac{1}{1 + e^{-2in_i}} \right) * \left[1 - \left(\frac{1}{1 + e^{-2in_i}} \right) \right]$$

$$S_1 = S_{in1} * \left(\frac{1}{1 + e^{-2in_1}} \right) * \left[1 - \left(\frac{1}{1 + e^{-2in_1}} \right) \right]$$

$\tau_{ii} = 1$

$$S_i = s_i n_i * z_i * (1 - z_i)$$

$$\begin{aligned} S_1 &= -0,018789 * 0,607052 * (1 - 0,607052) \\ &= -0,018789 * 0,238540 \\ &= -0,004482 \end{aligned}$$

$$\begin{aligned} S_2 &= -0,060300 * 0,501435 * (1 - 0,501435) \\ &= -0,060300 * 0,243368 \\ &= -0,014677 \end{aligned}$$

$$\begin{aligned} S_3 &= -0,044378 * 0,578232 * (1 - 0,578232) \\ &= -0,044378 * 0,243880 \\ &= -0,010823 \end{aligned}$$

$$\begin{aligned} S_4 &= -0,008240 * 0,556741 * 0,443256 \\ &= -0,008240 * 0,246780 \\ &= -0,002033 \end{aligned}$$

$$\Delta v_{ii} = a * S_i * X_{ii}$$

$$\Delta v_{11} = 1 * -0,004482 * 0,289518 = -0,001298$$

$$\Delta v_{12} = 1 * -0,014677 * 0,289518 = -0,004249$$

$$\Delta v_{13} = 1 * -0,010823 * 0,289518 = -0,003133$$

$$\Delta v_{14} = 1 * -0,002033 * 0,289518 = -0,000589$$

$$\Delta V_{2i} = a * S_i * X_{2i}$$

$$\Delta V_{21} = 1 * -0,0004482 * 0,337110 = -0,000151$$

$$\Delta V_{22} = 1 * -0,014677 * 0,337110 = -0,004948$$

$$\Delta V_{23} = 1 * -0,010823 * 0,337110 = -0,003649$$

$$\Delta V_{24} = 1 * -0,002033 * 0,337110 = -0,000686$$

$$\Delta V_{0i} = a * S_i * X_{0i}$$

$$\Delta V_{01} = 1 * -0,0004482 = -0,0004482$$

$$\Delta V_{02} = 1 * -0,014677 = -0,014677$$

$$\Delta V_{03} = 1 * -0,010823 = -0,010823$$

$$\Delta V_{04} = 1 * -0,002033 = -0,002033$$

$$V_{1i} = V_{11} + \Delta V_{1i}$$

$$V_{11} = 0,379729 + (-0,001298) = 0,378431$$

$$V_{12} = 0,309999 + (-0,004249) = 0,305750$$

$$V_{13} = 0,065896 + (-0,003133) = 0,062762$$

$$V_{14} = 0,11456 + (-0,000589) = 0,113971$$

$$V_{2i} = V_{22} + \Delta V_{2i}$$

$$V_{21} = 0,078035 + (-0,001511) = 0,076524$$

$$V_{22} = 0,244194 + (-0,004948) = 0,239246$$

$$V_{23} = 0,03013 + (-0,003649) = 0,009365$$

$$V_{24} = 0,385194 + (-0,0006) = 0,384594$$

$$V_{0i} = V_{0i} + \Delta V_{0i}$$

$$V_{01} = 0,298691 + (-0,00492) = 0,294210$$

$$V_{02} = 0,186595 + (-0,014677) = 0,141918$$

$$V_{03} = 0,292053 + (-0,010823) = 0,281230$$

$$V_{04} = 0,064919 + (-0,002033) = 0,062885$$

Rata - rata error Data 1 & Data 2

$$MSE = 0,093704 + 0,526702$$

2

$$= 0,620466 = 0,310233$$

2